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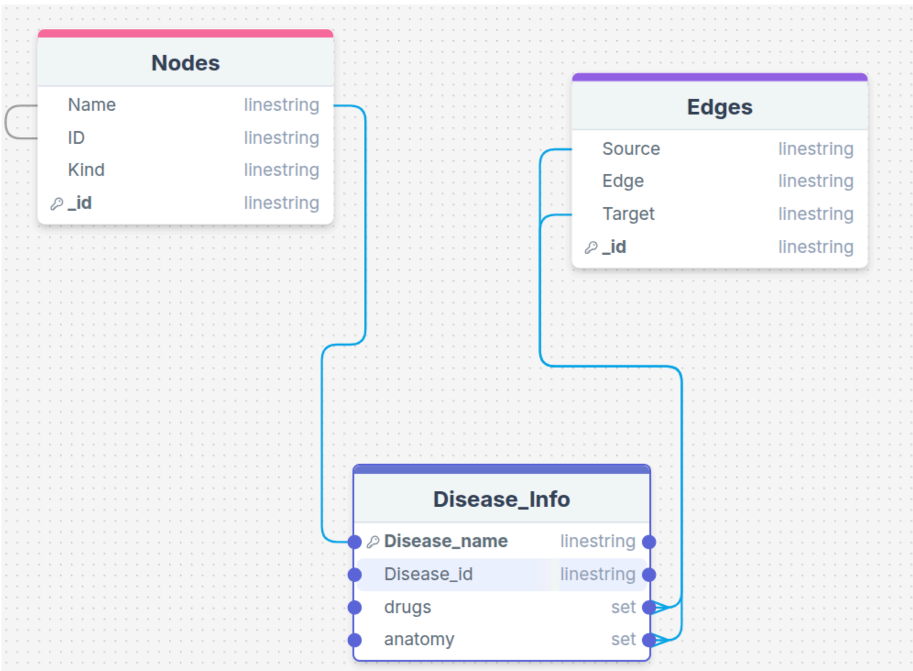
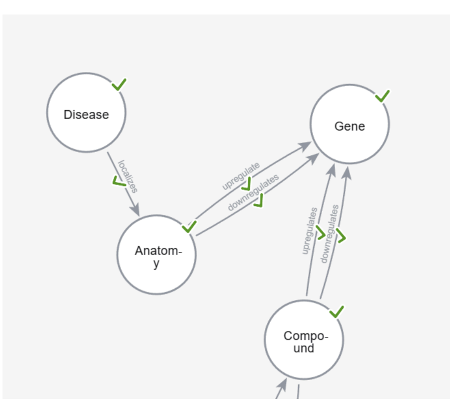
Big Data

Project #1



Query 1

I started by importing the data through python and reading csv. By doing so I was able to insert it into collections of nodes and edges. From there accessing the nodes and edges I was then able to use find_one on nodes to track down the nodes ID then from there from the doc I got the name from nodes. Then for the drug IDs I searched the edges for CpD and CtD using find, specifically bringing out source to find the compounds and then from there I looked up their names using the same method of finding the ID through source to then pull up the name in nodes. From there to find the anatomy of where it's localized I then used find looking for meta edges in edges that were DIA then from there using the target I used to find the ID in nodes to then pull out their names. From there I made another doc to bring forward the anatomy in Disease info. By creating each part into a variable I created a document that was inserted into a collection named Disease Info and at the same time printed the outcome of each part for the gui.



Query 2

I started by importing my data. I used the node.tsv to make the nodes: Disease, Anatomy, Gene, and Compound. For this, I filtered by kind. I then created only the necessary relationships for localizing, upregulating, and downregulating with the edges.tsv. For this, I filtered it with a metadata column. Then, I used my notes from class to write the query in cypher. Using the python driver, I wrote my function to input disease id and output all possible medications.

Optimization

While the current code is functional, its performance could be significantly improved, particularly with large datasets. The main areas for optimization involve reducing the number of database queries and enabling more efficient data retrieval. By indexing we would have been able to look up what was needed without looking through the whole database. We could have used mongodb pipeline where we could have made it so that the multiple queries used were instead joined. Specifically, the lookup could be used to perform joins, consolidating the multiple queries into a single, comprehensive database operation. This would fetch all the necessary disease, drug, and anatomy information in one go.

GUI

We ended up creating a simple terminal GUI.